

TASK 1 – Conditional Statements

1. Aim:-

To create a Python program that takes a student's marks as input and displays the grade using if, elif, and else statements.

2. Python Code:-

Task 1: Student Grade Calculator

new*

new*

new*

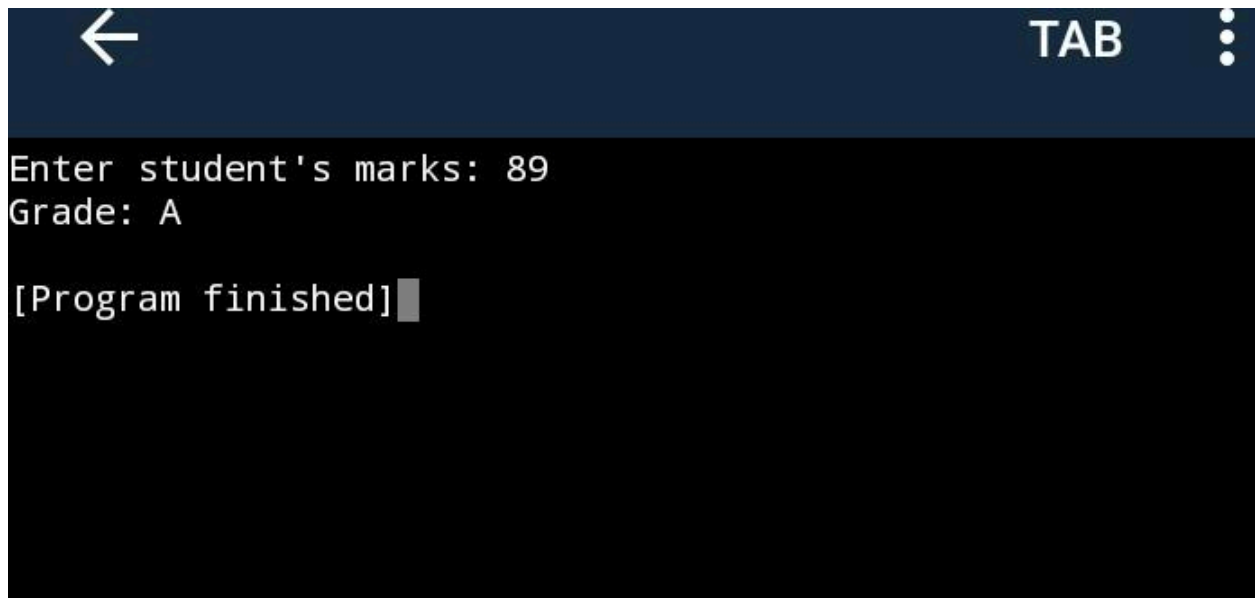
new*

new*

```
1
2 marks = float(input("Enter student's
3 marks: "))
4 if marks < 0 or marks > 100:
5     print("Invalid marks! Please enter mark
6     between 0 and 100.")
7 elif marks >= 90:
8     print("Grade: A+")
9
10 elif marks >= 80:
11     print("Grade: A")
12
13 elif marks >= 70:
14     print("Grade: B")
15
16 elif marks >= 60:
17     print("Grade: C")
18
19 elif marks >= 50:
20     print("Grade: D")
21
22 else:
23     print("Grade: Fail")
```

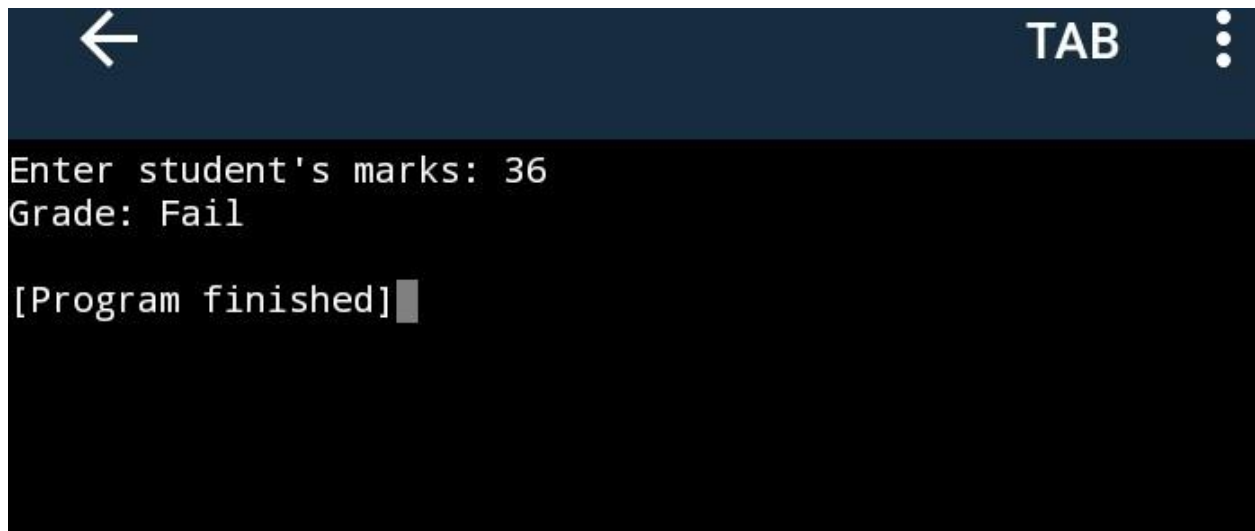
3. Output:-

#1)



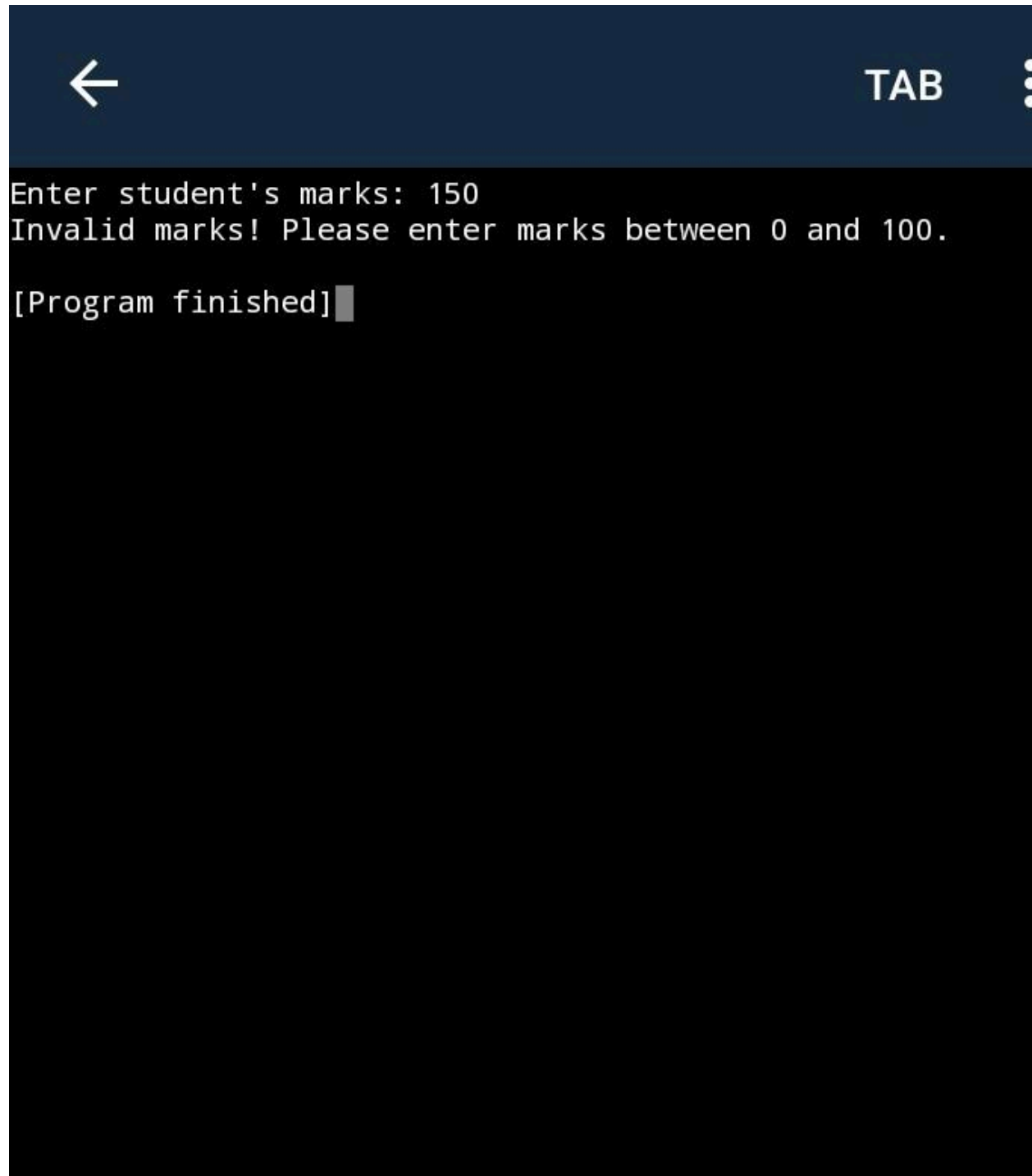
A screenshot of a mobile application interface. The top bar is dark blue with a white back arrow on the left, the text 'TAB' in the center, and a white three-dot menu icon on the right. The main content area is black with white text. It displays 'Enter student's marks: 89' and 'Grade: A' on two lines. Below this, it shows '[Program finished]' followed by a small grey rectangular cursor.

#2)



A screenshot of a mobile application interface, similar to the first one. The top bar is dark blue with a white back arrow on the left, the text 'TAB' in the center, and a white three-dot menu icon on the right. The main content area is black with white text. It displays 'Enter student's marks: 36' and 'Grade: Fail' on two lines. Below this, it shows '[Program finished]' followed by a small grey rectangular cursor.

#3)



```
← TAB
Enter student's marks: 150
Invalid marks! Please enter marks between 0 and 100.
[Program finished]
```

Explanation:

This program accepts the student's marks as input and uses if, elif, and else conditional statements to determine the grade. It also checks whether the entered marks are valid. Marks should be between 0 and 100.

TASK 2 – Loops:-

1. Aim:-

To create a Python program using loops to display numbers from 1 to N, N to 1, and a multiplication table.

2. Python Code:-

new*

new*

new*

new*

new*

```
1
2 n = int(input("Enter a number: "))
3
4 print("\nNumbers from 1 to N:")
5 for i in range(1, n + 1):
6     print(i, end=" ")
7
8 print("\n\nNumbers from N to 1:")
9 for i in range(n, 0, -1):
10     print(i, end=" ")
11
12 print("\n\nMultiplication Table:")
13 for i in range(1, 11):
14     print(n, "x", i, "=", n * i)
```

Output:-

```
← TAB ⋮
Enter a number: 5
Numbers from 1 to N:
1 2 3 4 5
Numbers from N to 1:
5 4 3 2 1
Multiplication Table:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
[Program finished]
```

Explanation:

This program uses a for loop to print numbers from 1 to N, another for loop to print numbers from N to 1, and a third loop to display the multiplication table from 1 to 10.

TASK 3 – Functions

1. Aim:-

To create a simple calculator using user-defined functions for addition, subtraction, multiplication, division, and modulus.

2. Python Code:-

new*

new*

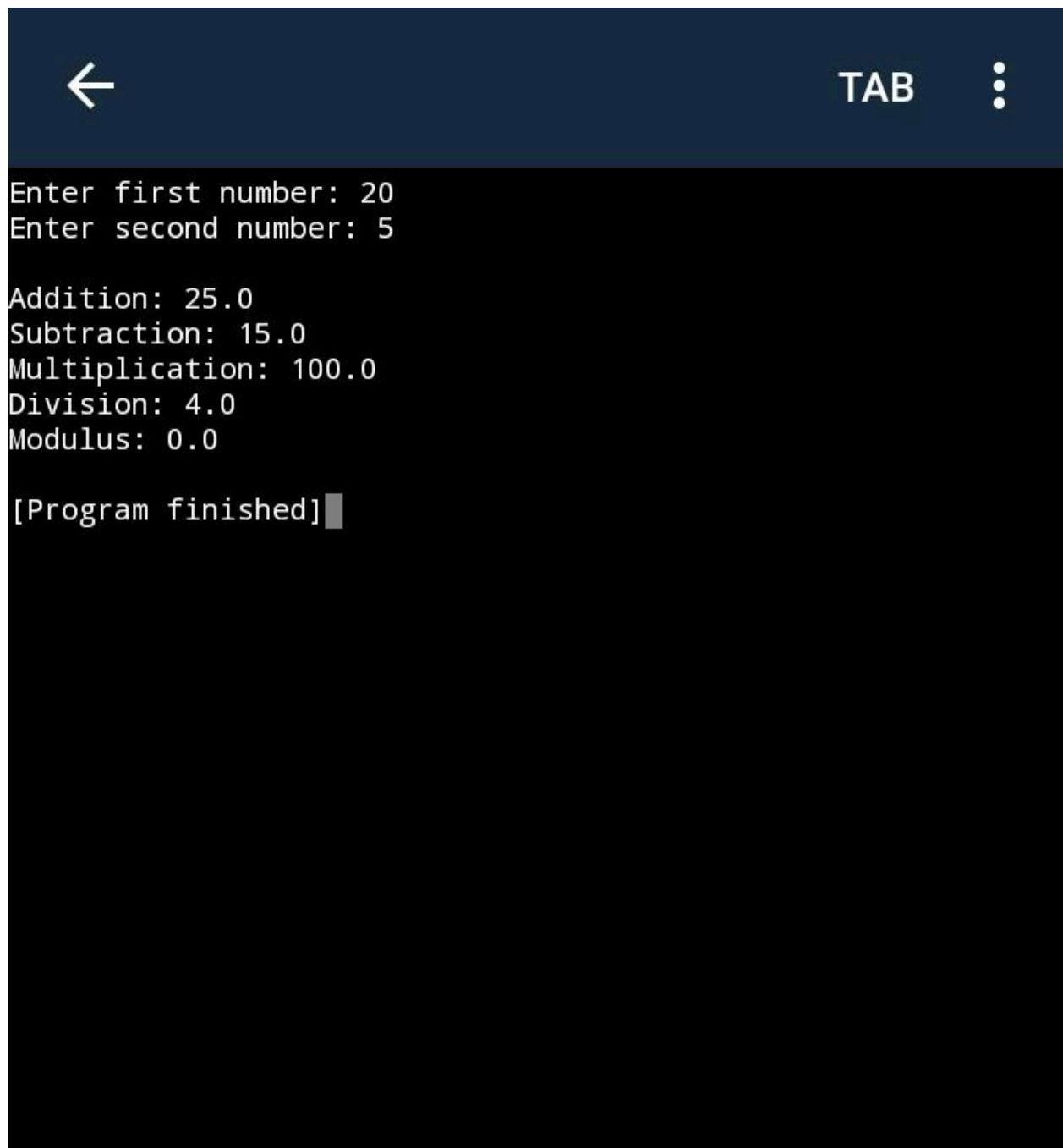
new*

new*

new*

```
1 def add(a, b):
2     return a + b
3
4 def subtract(a, b):
5     return a - b
6
7 def multiply(a, b):
8     return a * b
9
10 def divide(a, b):
11     return a / b
12
13 def modulus(a, b):
14     return a % b
15
16
17 a = float(input("Enter first number: "))
18 b = float(input("Enter second number: "))
19
20 print("\nAddition:", add(a, b))
21 print("Subtraction:", subtract(a, b))
22 print("Multiplication:", multiply(a, b))
23
24 if b != 0:
25     print("Division:", divide(a, b))
26     print("Modulus:", modulus(a, b))
```

Output:-

A screenshot of a mobile application interface. At the top, there is a dark blue header bar with a white back arrow on the left, the word 'TAB' in white in the center, and a white three-dot menu icon on the right. Below the header, the main area has a black background with white text. The text displays the results of arithmetic operations: 'Enter first number: 20', 'Enter second number: 5', 'Addition: 25.0', 'Subtraction: 15.0', 'Multiplication: 100.0', 'Division: 4.0', 'Modulus: 0.0', and '[Program finished]' followed by a small grey square cursor.

```
Enter first number: 20
Enter second number: 5

Addition: 25.0
Subtraction: 15.0
Multiplication: 100.0
Division: 4.0
Modulus: 0.0

[Program finished]
```

Explanation:

This program uses five user-defined functions: `add()`, `subtract()`, `multiply()`, `divide()`, and `modulus()`. Each function performs a different arithmetic operation. The program takes two numbers from the user and displays the results.

TASK 4 – Scope of Variables:-

1. Aim:-

To demonstrate the use of local and global variables in Python.

2. Python Code:-

new*

new*

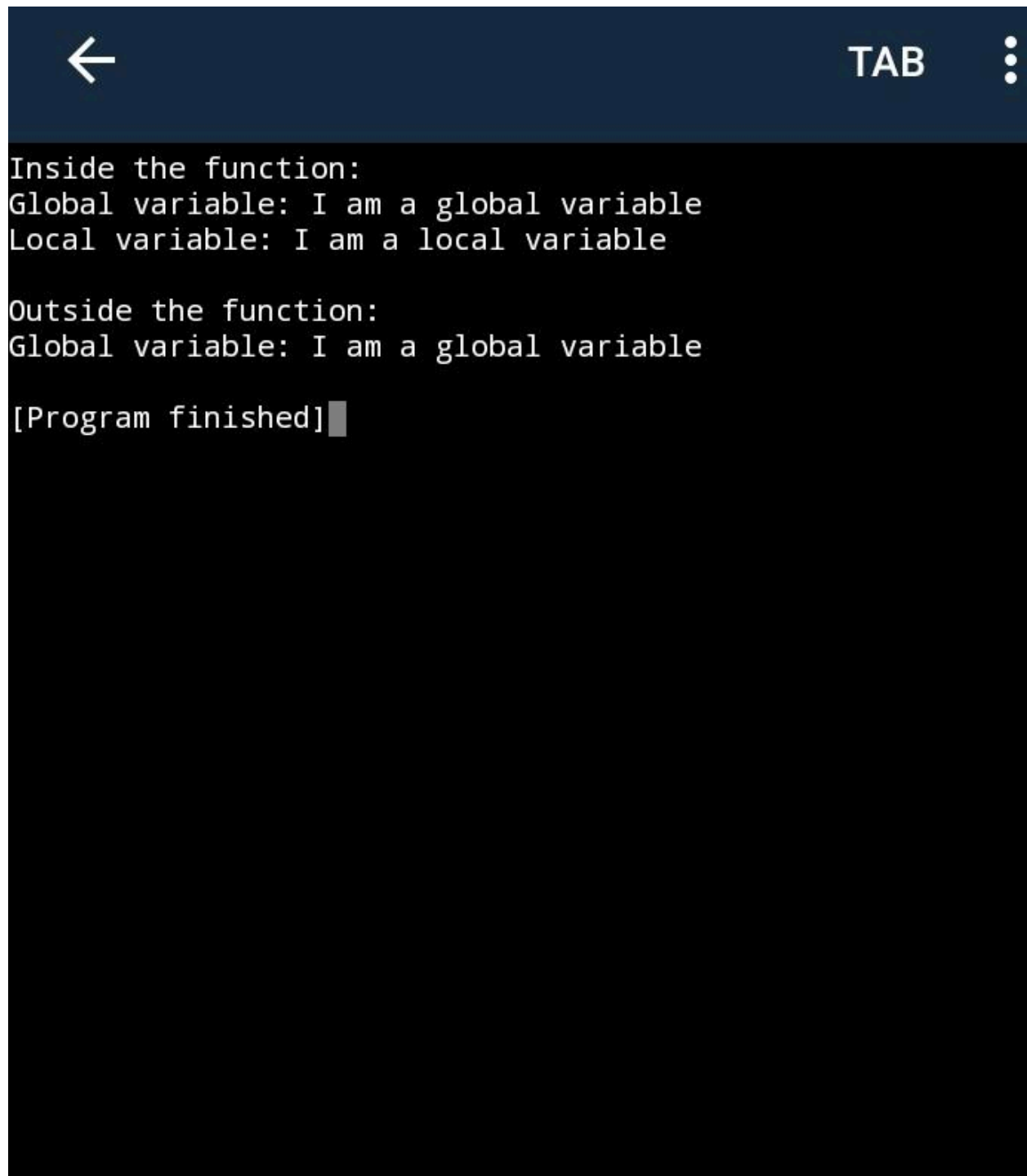
new*

new*

new*

```
1 message = "I am a global variable"
2
3 def display_message():
4     # Local variable
5     local_message = "I am a local variable"
6
7     print("Inside the function:")
8     print("Global variable:", message)
9     print("Local variable:", local_message)
10
11
12 display_message()
13
14 print("\nOutside the function:")
15 print("Global variable:", message)
```

Output:-

A screenshot of a terminal window with a dark blue header bar. The header bar contains a white back arrow icon on the left, the text 'TAB' in the center, and a white three-dot menu icon on the right. The main area of the terminal is black with white text. The text displayed is: 'Inside the function:', 'Global variable: I am a global variable', 'Local variable: I am a local variable', 'Outside the function:', 'Global variable: I am a global variable', and '[Program finished]' followed by a small grey cursor block.

```
← TAB ⋮  
Inside the function:  
Global variable: I am a global variable  
Local variable: I am a local variable  
  
Outside the function:  
Global variable: I am a global variable  
  
[Program finished]
```

4. Explanation:-

A global variable is declared outside a function and can be accessed throughout the program. A local variable is declared inside a function and can be accessed only within that function.

TASK 5 – Modules:-

1. Aim:-

To demonstrate the use of built-in Python modules such as math, random, and datetime.

2. Python Code:-

new*

new*

new*

new*

new*

```
1 import math
2 import random
3 import datetime
4
5 # Using math module
6 number = 25
7 print("Square root of", number, ":", math.
  sqrt(number))
8
9 # Using random module
10 print("Random number between 1 and
   100:", random.randint(1, 100))
11
12 # Using datetime module
13 current_date = datetime.date.today()
14 print("Today's date:", current_date)
```

Output:-



TAB



```
Square root of 25 : 5.0  
Random number between 1 and 100: 78  
Today's date: 2026-08-09  
  
[Program finished]
```

Explanation:

This program demonstrates three built-in Python modules. The math module is used to calculate the square root, the random module generates a random number, and the datetime module displays the current date.
